Twist, Interpolation and Resolution Limits of Curvature-Based Shape Sensing for Soft Continuum Arms

Anthony Ijoot
Department of Electrical and Electronics Engineering
Makerere University, Kampala, Uganda
anthonyijoot98@gmail.com

***Abstract*—Soft continuum arms often estimate their shape by integrating curvature measured by a few embedded sensors, yet the number of sensors, the components they measure, the way their readings are interpolated and the resolution of their converters are usually chosen by trial on a prototype. We study these choices in simulation, where the true shape is known, for a three-segment, 25 cm elastomer arm under gravity and tip loads. A static Kirchhoff rod model, cross-checked against the PyElastica Cosserat rod simulator in tip position and twist, gives 400 equilibrium shapes; sensors with a 10 mm gauge, noise and quantization are placed along them, and the shape is rebuilt by integrating the measured curvature. When twist is not measured and is taken as zero, as kinematic reconstruction usually assumes, the median tip error stays near 1.9 % of the arm length (6.1 % at the 95th percentile) however many bending sensors are used, because gravity and loads twist the arm; for a torsionally soft arm it reaches 7.9 %. Measuring twist lowers it to 0.10 % with 12 sensors. The constant-curvature-per-segment model stops at 0.6 % even with twist, and interpolating within the actuated segments rather than across their boundaries reduces it by a third to a half at 6 and 12 sensors. An error model predicts the effect of noise and quantization to within 20 % and shows that 11-bit conversion suffices when the sensors' own noise is below 0.006 m^{-1}; noisier sensors need fewer bits.**

***Index Terms*—soft robotics, continuum robots, proprioception, shape sensing, Cosserat rod, curvature sensors, twist, quantization**

I. Introduction

A soft continuum arm has no joints to encode. Its configuration is a curve, and closed-loop control, contact estimation and teleoperation all need an estimate of that curve from sensors carried by the arm itself [1]. Optical fibres with Bragg gratings, embedded resistive or capacitive strain gauges and liquid-metal or ionic sensors all report local strain, which a bending model turns into curvature [2], [3]. The shape then follows by integrating the curvature along the arm, the kinematics behind the constant-curvature models of continuum robots [4]. Learned maps from many soft sensors to configuration [5], [6] and probabilistic estimators that fuse a mechanical model with the measurements [7] are alternatives, but they too depend on what the sensors can observe.

The designer of the sensing hardware faces coupled choices: how many sensors to embed, which components of curvature they must measure, how to interpolate between them, and how finely to digitize them. Each sensor costs wiring, stiffness and an analog channel. Twist is a known difficulty in fibre shape sensing, where measuring it needs helical or specially arranged gratings [8], but its effect on the shape of a soft arm loaded by its own weight is rarely quantified, and the other choices are usually settled on a prototype.

This paper studies these choices in simulation. Its contributions are: a static rod model of a three-segment arm under gravity and tip loads, cross-checked against a second simulator, used to produce 400 equilibrium shapes; a sensing model with gauge averaging, noise and quantization, and three reconstructions; the finding that twist, not sensor count, limits kinematic reconstruction from bending-only sensors, and that the limit grows as the arm becomes softer in torsion; a comparison of interpolation within and across actuated segments; and an error model that gives the converter resolution worth using for a given sensor count and noise.

II. Arm Model

A. Static Kirchhoff rod

Following the rod models used for continuum robots [9], [10], the arm is an inextensible, unshearable rod of length L, clamped at the base with its tangent horizontal. With arc length s, centreline $p(s)$, material frame $R(s)$ and body-frame curvature $u(s) = (u_1, u_2, u_3)$, where u_3 is the twist of the material frame about the centreline,

$$p' = R\,e_3, \qquad R' = R\,\hat{u}, \tag{1}$$

with $\hat{u}$ the skew matrix of u. The internal force n and moment m, in the spatial frame, balance the distributed weight f:

$$n' + f = 0, \qquad m' + p' \times n = 0. \tag{2}$$

Actuation is modelled as an intrinsic bending curvature $u^*(s)$ that is constant within each segment, as produced by pressurizing a segment's chambers or pulling its tendons, with no intrinsic twist:

$$R^\top m = K\,(u - u^*), \qquad K = \mathrm{diag}(EI, EI, GJ). \tag{3}$$

The tip carries a force F and no moment. The unknown base force and moment are found by Newton's method on the tip conditions, integrating (1)–(3) with 300 fourth-order Runge-Kutta steps and ramping actuation and load in three steps. Each shape takes about 1.5 s.

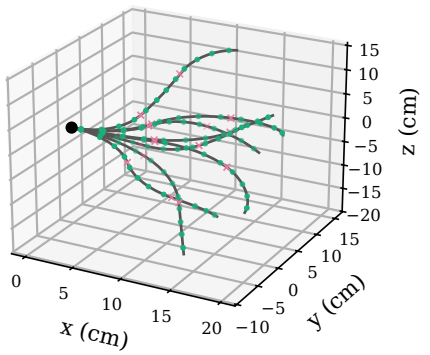


Fig. 1. Eight of the 400 equilibrium shapes. Base at the origin, tangent along x, gravity along $-z$. Dots: 12 sensor positions; crosses: segment boundaries.

B. Parameters and cross-checks

The arm is 0.25 m long with a solid 15 mm radius, density 1080 kg/m^3 and an effective modulus of 3 MPa, typical of a fibre-reinforced elastomer; its own weight bends it 30 mm downward when unactuated. Three segments of equal length are actuated. For a solid incompressible section $GJ/EI = 2/3$; Section IV-A varies this ratio.

The solver reproduces a circular arc exactly under pure intrinsic curvature, and for a stiff rod under its own weight it matches the linear Euler-Bernoulli tip deflection to five digits. It was also compared with PyElastica [11], a dynamic Cosserat rod simulator that includes shear and extension, damped to rest. The tip positions agree to 0.09 mm under gravity alone. With three-dimensional actuation, and with a tip load added, they differ by 1.7 and 1.2 mm (under 0.7 % of L) with 240 PyElastica elements; in the first case the difference is 6.5, 3.3 and 1.7 mm with 60, 120 and 240 elements, halving with each refinement as the curvature steps at segment boundaries are resolved more finely. The twist, the quantity central to this paper, agrees as well: with the tip load the largest u_3 is 1.064 against 1.050 m^{-1}, the rms difference along the arm is 0.002 m^{-1}, and the total twist is $-1.64°$ against $-1.58°$. The sensing study compares reconstructions with shapes of the same rod model, so its conclusions do not rest on the residual differences.

C. Shape data set

Each of 400 shapes draws, for every segment, a bending direction uniformly in $[0, 2\pi)$ and an intrinsic curvature magnitude uniformly up to a 90° bend of the segment (18.8 m^{-1}). With probability one half, 213 of the 400 shapes, the tip also carries a force of random direction and magnitude up to 0.3 N, a sixth of the arm's weight, representing contact or a payload. The median tip displacement from the straight arm is 0.68 L (Fig. 1). The largest bending curvature is 19.2 m^{-1}, and although actuation applies no twist, gravity and the loads twist the arm by up to 2.3 m^{-1}.

III. Sensing and Reconstruction

A. Sensor model

Sensor j, centred at $s_j = (j + \frac{1}{2})L/N$ with gauge length $g = 10\,\mathrm{mm}$, reports the exact average of the two bending curvatures (u_1, u_2) over its gauge; a three-axis sensor also reports the twist u_3. White noise of rms σ is added to every channel, and the result is quantized with B bits over $\pm\kappa_{\mathrm{fs}} = \pm 20\,\mathrm{m}^{-1}$, which covers every curvature in the data set. For sensor counts that are multiples of three, every segment holds $N/3$ sensors and none sits on a segment boundary.

B. Reconstruction

The estimated curvature $\hat{u}(s)$ is integrated through (1) from the clamped base. When twist is not measured it is set to zero, which treats the sensors' frame as one that does not rotate about the centreline, the usual assumption of kinematic shape reconstruction. Each integration step applies the exact rotation of its constant curvature. Three estimates of $\hat{u}$ are compared, all from the same sensor positions:

- *one curvature per segment*: the mean of the segment's sensors, held over the segment, the piecewise constant-curvature model [4];
- *across segments*: linear interpolation between neighbouring sensors along the whole arm, held constant beyond the end sensors;
- *within segments*: linear interpolation between the sensors of a segment only, held constant from the outermost sensor to the segment boundary, so that changes of actuation at the boundaries are not smeared.

The last needs the segment boundaries, which the designer knows. Integrating the true curvature, twist included, reproduces the true shape to 0.07 % of L at worst, the numerical floor of the comparison. Errors are reported as the tip position error divided by L, as the median and 95th percentile over the 400 shapes.

IV. Results

A. Twist limits bending-only reconstruction

Fig. 2 shows the tip error of ideal sensors against their number. With bending-only sensors every reconstruction levels off at a median of 1.9 to 2.2 % of L (95th percentile 6.0 to 6.3 %), from 6 sensors up to 24. The cause is twist. Gravity acting on a three-dimensionally bent arm, and any lateral tip force, twist the arm about its own axis; a sensor fixed to the material then reports its curvature in a frame that has rotated, and a reconstruction that ignores the rotation bends the wrong way. Ignoring twist alone, with the bending curvature known exactly everywhere, already produces a median of 1.9 %. Twist is not caused by the tip loads alone: without them the median is 1.8 % and the 95th percentile 6.2 %. Measuring twist removes the floor: with 12 sensors the median falls to 0.10 % and the 95th percentile to 0.17 %, about 19 and 36 times lower.

The floor depends on how easily the arm twists (Table I). With $GJ/EI = 0.2$, as in a thin-walled or multi-chamber section, the bending-only median rises to 7.9 % and the 95th

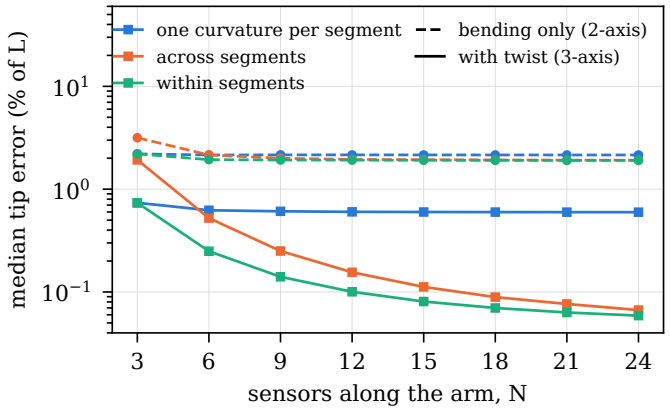


Fig. 2. Median tip error against the number of ideal sensors, for the three reconstructions, with bending-only (dashed) and three-axis (solid) sensors.

TABLE I
TIP ERROR (% OF L, MEDIAN / 95TH PERCENTILE) WITH 12 IDEAL SENSORS AND INTERPOLATION WITHIN SEGMENTS, AGAINST THE RATIO OF TORSIONAL TO BENDING STIFFNESS

GJ/EI	Shapes	Max twist (m^{-1})	Bending only	With twist
0.2	60	8.3	7.89 / 20.27	0.19 / 0.58
0.67	400	2.3	1.92 / 6.10	0.10 / 0.17
2.0	60	0.8	0.79 / 2.09	0.08 / 0.10

percentile to 20 %; a torsionally stiff, fibre-wound arm with $GJ/EI = 2$ still shows 0.8 %. Three-axis sensing stays below 0.6 % at the 95th percentile in all three cases. The two extra rows use 60 new random shapes each, so they compare trends rather than identical poses.

The floor is a limit of kinematic integration, not of the information the arm carries. The twist here comes from the torsional moment of gravity and the tip load, so an estimator that combines the bending measurements with a mechanical model of the arm and its loads [7] could predict much of it. Such an estimator needs the model and the loads; a twist measurement needs neither.

B. Constant curvature per segment is not enough

With twist measured, one curvature per segment reaches 0.74 % with three sensors and stops near 0.6 % however many sensors are averaged (Fig. 2). Under gravity and tip loads the curvature varies along each segment, because the bending moment of the weight and the load grows towards the base, and a constant-curvature model cannot represent the variation.

C. Interpolating within segments

Linear interpolation follows the variation, but it matters where it is allowed to act. Interpolating across a segment boundary smears the change of actuation there over a sensor spacing; interpolating within segments keeps it. From the same sensor positions the median error falls from 0.52 to 0.25 % with 6 sensors, from 0.25 to 0.14 % with 9 and from 0.16 to 0.10 % with 12; beyond 15 sensors both approach about 0.06 %. The benefit does not depend on the actuation changing abruptly: when the intrinsic curvature changes linearly over 20 mm at each boundary (200 further shapes), the median falls from 0.46 to 0.28 % with 6 sensors and from 0.19 to 0.11 % with 12, though hardly at all with 9 or 24.

D. Gauge length

The gauge length hardly matters. From 2 to 20 mm the median error with 12 three-axis sensors stays between 0.09 and 0.10 %, and with 6 sensors between 0.23 and 0.25 %: within a segment the curvature varies slowly enough that a long stretchable gauge loses nothing against a short one.

E. Resolution and noise

A curvature error held over the sensor spacing L/N at arc length s tilts the rest of the arm and moves the tip by about $(L - s)\,L/N$ times the error. If each sensor carries an independent error of rms σ_e on each bending axis, summing over sensors and two axes gives

$$\frac{e_{\mathrm{rms}}}{L} \approx \sqrt{\left(\frac{e_0}{L}\right)^2 + \frac{2}{3}\,\frac{\sigma_e^2 L^2}{N}}, \qquad \sigma_e^2 = \sigma^2 + \frac{q^2}{12}, \tag{4}$$

where $q = 2\kappa_{\mathrm{fs}}/2^B$ is the quantization step and e_0 the error with ideal sensors, taken from the simulation. The model neglects the error on the twist channel and uses the straight-arm lever arm $L-s$, which overstates the lever of a curved arm; the two partly cancel. Fig. 3 compares (4) with the simulated rms error for 6, 12 and 24 sensors. At 8 and 10 bits, and for white noise of 0.05 and 0.2 m^{-1}, the model is within 20 %. At 6 bits with 24 sensors it underestimates the error by a third: the gauges then nearly touch, neighbouring sensors see almost the same curvature, and their quantization errors are no longer independent.

The quantization term halves with each bit while e_0 falls only as sensors are added, so each sensor count has a resolution beyond which bits are wasted. Table II lists the fewest bits that keep the 95th percentile within 20 % of its ideal-sensor value for noise-free sensors: 8 bits for three sensors, 10 for six to nine and 11 for twelve or more. At 8 bits, 12 sensors give a median of 0.24 %, 2.4 times their ideal value. These counts hold only while the sensors' own noise is below the quantization noise, about 0.006 m^{-1} at 11 bits. Real soft sensors are often noisier: white noise of 0.05 m^{-1} rms, a quarter of a percent of the full-scale value, costs as much as 8-bit quantization (a median of 0.24 % with 12 sensors). With such sensors, (4) shows that bits beyond the point where $q/\sqrt{12}$ falls below σ buy little, and lowering the noise or adding sensors is what helps.

V. DISCUSSION

Design guidance. For an arm of this kind the results suggest an order of priorities. First, account for twist: measure it, for instance with helically arranged gauges or gratings [8], build the arm stiff in torsion, or estimate it with a model of the arm and its loads. Second, interpolate within the actuated segments

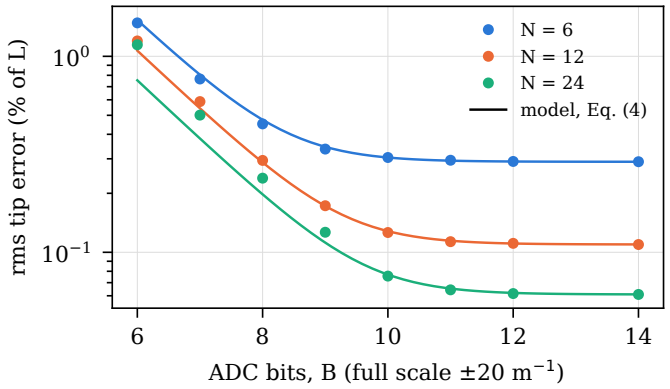


Fig. 3. Rms tip error against converter resolution for noise-free three-axis sensors interpolated within segments. Dots: simulation; lines: the model (4).

TABLE II
FEWEST BITS KEEPING THE 95TH-PERCENTILE TIP ERROR WITHIN 20 % OF THE IDEAL-SENSOR VALUE (NOISE-FREE THREE-AXIS SENSORS, INTERPOLATION WITHIN SEGMENTS, FULL SCALE ±20 m^{-1})

Sensors N	Ideal 95th pct. (% of L)	Bits	95th pct. at those bits (% of L)
3	1.33	8	1.45
6	0.46	10	0.50
9	0.25	10	0.27
12	0.17	11	0.17
18	0.11	11	0.12
24	0.09	11	0.10

rather than across their boundaries, and avoid the constant-curvature model when gravity or loads are significant. Third, choose the converter from (4) and the measured sensor noise. With twelve noise-free three-axis sensors at 11 bits the tip stays within 0.17 % of the length, 0.4 mm on this arm, for 95 % of the shapes.

Scope. The study is a simulation of one arm. The rod is static, inextensible and isotropic in bending; real elastomer arms also extend under pressure, may couple actuation to twist, creep and show hysteresis, and real sensors drift with temperature and couple bending to strain. A real twist channel is likely to be noisier than a bending channel, which (4) does not include. The tip loads are applied forces rather than contact, and dynamics are not considered. The numbers therefore describe how the error scales with the sensing choices rather than the accuracy of a particular prototype; the next step is to measure the predicted twist floor and interpolation gain on a fabricated arm with motion capture.

VI. CONCLUSION

Kinematic reconstruction of a soft arm's shape from embedded curvature sensors is limited by what the sensors miss before it is limited by how many there are. Bending-only sensors leave errors of about 2 % of the length on a solid elastomer arm and 8 % on a torsionally soft one, because gravity and loads twist the arm. Measuring twist lowers the error by more than an order of magnitude, interpolating within actuated segments lowers it by a further third to a half at small sensor counts, and a simple error model gives the converter resolution beyond which, for a given sensor noise, more bits buy nothing.

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